

PROJECT MILESTONE #1

Title: Fairness or Bias of Large Language Models (LLMs) in High-Stake Domains (Hiring and Recidivism)

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Research Problem: Large Language Models (LLMs) such as GPT and BERT are increasingly being used in tasks such as hiring and criminal risk assessment due to their benefit of being able to handle large amounts of data. However, since such models are trained on human data, they inevitably inherit social, gender, and racial biases. This project aims to examine the fairness of the LLMs used in such high-stakes environments.

Background: Algorithmic bias has been an issue for several years in algorithmic decision-making. Some popular examples are that the COMPAS recidivism prediction algorithm has faced criticism for exhibiting racial bias, while automated resume screening systems have mirrored gender bias because of biased training data. As LLMs become more advanced, the problem is exacerbated.

Research Questions:

1. How do LLMs show bias when used in hiring or recidivism prediction?
2. What current methods are employed to identify or minimize these biases?
3. How can we measure how fair LLM is in high-stakes situations?

General Approach/Methodology: We are going to adopt a literature review approach to analyze the existing studies focusing on LLM bias and fairness issues in the criminal justice system and employment. We will assess individual papers on the applied bias detection methods and the proposed strategy for mitigation. Our goal in this project is to provide an overview of dominant results, recognize recurring patterns across the research and emphasize research areas that need additional investigation.

Evaluation Metrics: We will take into account some fairness indicators, which include:

- **Demographic Parity/Statistical Parity Difference:** Ensuring every group gets a fair opportunity for a favorable outcome.
- **Equal Opportunity/Equalized Odds:** Maintaining similar true positive and false positive rates throughout the group.
- **False Positive and False Negative Rate Balance:** Reducing systematic error in prediction.
- **Disparate Impact Ratio:** proportional fairness testing.

By synthesizing results from multiple studies, this project will offer recommendations on best practices in measuring and maximizing fairness in LLMs. We hope that in the end, the results

will promote clear and ethical use of AI in situations where algorithmic results have a direct impact on human life.

List of 10 Papers for Literature Survey:

1. Gaebler, J. D., Goel, S., Huq, A., & Tambe, P. Auditing the use of language models to guide hiring decisions (2024). URL <https://arxiv.org/abs/2404.03086>.
2. Wilson, K., & Caliskan, A. (2024, October). Gender, race, and intersectional bias in resume screening via language model retrieval. In Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (Vol. 7, pp. 1578-1590).
3. Glazko, Kate, et al. "Identifying and improving disability bias in GPT-based resume screening." Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency. 2024.
4. Anzenberg, E., Samajpati, A., Chandrasekar, S., & Kacholia, V. (2025). Evaluating the Promise and Pitfalls of LLMs in Hiring Decisions. arXiv preprint arXiv:2507.02087.
5. Rao, P. S., Venkatesan, L. N., Cherubini, M., & Jayagopi, D. B. (2025). Invisible Filters: Cultural Bias in Hiring Evaluations Using Large Language Models. arXiv preprint arXiv:2508.16673.
6. Wilson, Kyra, et al. "No Thoughts Just AI: Biased LLM Hiring Recommendations Alter Human Decision Making and Limit Human Autonomy." Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society. Vol. 8. No. 3. 2025.
7. Xu, Jiannan, Gujie Li, and Jane Yi Jiang. "AI Self-preferencing in Algorithmic Hiring: Empirical Evidence and Insights." Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society. Vol. 8. No. 3. 2025.
8. Veldanda, A. K., Grob, F., Thakur, S., Pearce, H., Tan, B., Karri, R., & Garg, S. (2023, November). Investigating hiring bias in large language models. In *R0-FoMo: Robustness of Few-shot and Zero-shot Learning in Large Foundation Models*.
9. Cohen, Lee, et al. "Two Tickets are Better than One: Fair and Accurate Hiring Under Strategic LLM Manipulations." arXiv preprint arXiv:2502.13221 (2025). URL <https://arxiv.org/abs/2502.13221>.
10. Ding, Lei, et al. "Probing Social Bias in Labor Market Text Generation by ChatGPT: A Masked Language Model Approach." Advances in Neural Information Processing Systems 37 (2024): 139912-139937.